PROJECT TRAINEE REPORT ON E-COMMERCE

"Full Stack Development-J2EE & Web 2.0 Technologies"

Project done at Vizag Steel Plant, Visakhapatnam

A project report submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY (B.Tech.)

COMPUTER ENGINEERING

by

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NATIONAL INSTITUTE OF ADVANCED MANUFACTURING TECHNOLOGY

Under the Esteemed Guidance of

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VIZAG STEEL PLANT

Visakhapatnam, Andhra Pradesh – 530031

CERTIFICATE OF APPROVAL

This is to certify that we have examined the project trainee report entitled **"Full Stack Web Development"**,

submitted by the following group of students during their internship at **Visakhapatnam Steel Plant (RINL)**:

Name	Trainee No.	College & Place
Kunja Rahul	100049220	NIAMT, RANCHI

I hereby accord our approval of this work, which has been presented in a manner suitable for its acceptance towards the partial fulfillment of the requirements for the **Bachelor of Technology (B.Tech)** degree.
This approval does not necessarily endorse every statement, opinion, or conclusion presented in the report. It signifies that the report meets the academic criteria for submission.

Project Guide

(MS. SHRIYA BASUMALLIK)

Manager

IT & ERP Department

Vizag Steel Plant

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ACKNOWLEDGEMENT

I would like to express our sincere gratitude to **Rashtriya Ispat Nigam Limited (Visakhapatnam Steel Plant)** for giving us the opportunity to undertake this valuable project-based internship on **Full Stack Web Development**.

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Finally, I acknowledge the cooperation and support extended by the IT & ERP department staff, which made my internship a productive and enriching experience.

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INDEX

S.No	Content	Page No.
1.	About RINL	5-6
2.	About ERP	7
3.	About Full Stack	8-9
4.	Frontend Development	10-18
5.	Backend Development	19-23
6.	Database Integration	24-27
7.	Introduction	28
8.	Objectives	29
9.	Key features	30
10.	Technology Stack	31-32
11.	System Architecture	33
12.	ER Diagram	34-35
13.	Modules	36
14.	Database Design	37-38
15.	Code Snippets	39-42
16.	Stylesheets & UI Design	43-47
17.	Interface Overview & Page Layout	48-49
18.	Security and Considerations	49
19.	Testing and Debugging	50
20.	Future Scope	50
21.	References	51
22.	Conclusion	52-54

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ABOUT RINL

Rashtriya Ispat Nigam Limited, the corporate entity of Visakhapatnam Steel Plant is a Navaratnam PSE under the Ministry of Steel. Visakhapatnam Steel Plant fondly called Vizag steel. It is the first shore based Integrated Steel Plant in the country and is known for its quality products delighting the customers. It is a market leader in long products, and it caters to the needs of diverse Industrial sectors. It is the first Steel plant to be certified ISO 9001:2008 (presently2015), ISO 14001:2004 (presently2015), OHSAS 18001:2007 and ISO/IEC 27001:2013 Standards. It is also the first PSE to be certified ISO 50001:2011 - Energy Management Systems and has acquired CMMI Level 3 Certification for s/w development.

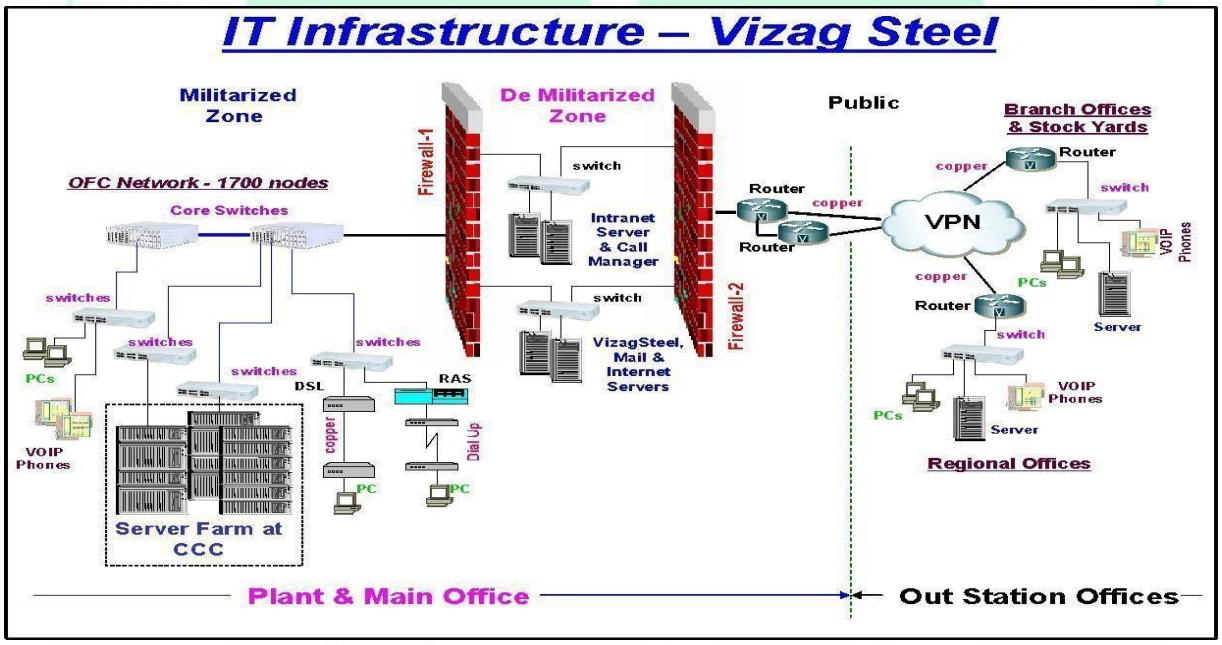
Major Technologies / Facilities Deployed
Open Systems RISC Servers: Windows servers
Optical Fiber Cabling spanning the labyrinth of departments & township
TCP/IP based Gigabit Ethernet Network
Windows based workstations as clients
Client-Server Architecture
Oracle 11g/12c as Database
Forms-12c, Reports-12c as application development tools
Oracle-12c 3-tier Application Server architecture for browser-based application access
GUI based applications that provides ease of data entry & editing
Intranet / Internet Applications using .net / J2EE

Major Business Applications Systems
Marketing systems
Human Resource Information Systems
Payroll System
Production Planning and Control System
Maintenance Management System
Financial account System
Materials Management System

ABOUT ERP

Enterprise Resource Planning (ERP) is one of the major set of modules of enterprise systems being implemented in various organizations across the globe. ERP seeks to standardize, streamline, and integrate the various operations and information flows in a company by synergizing its resources. ERP is one of the most advanced business management applications available to support most of the business processes of manufacturing as well as service companies in a wide variety of industries.

ERP systems provide an integrated application environment with a fast and seamless access to single unified information business-wide, hence representing a catalyst for business process change.



Critical Success Factors for ERP implementation:

- Undiluted management commitment
- Business driven initiative
- Experienced consultants
- Cross functional participation in core team
- Strong leadership with clear vision
- Predefined To-Be processes
- Implementation cycle to be of proper duration
- Proper Change management

➢ Introduction to Full Stack Web Development

❖ What is Full Stack Web Development?

- **Full Stack Web Development** refers to the development of both the **frontend (client-side)** and **backend (server-side)** components of a web application. A full stack developer is responsible for designing how the application looks and behaves in the browser, as well as building the backend logic, server-side APIs, database operations, and overall system flow.
- In this project, we used **HTML, CSS, and JavaScript** to build the **frontend**, while the **backend** was developed entirely in **Java using Spring Boot (Maven project)**. The backend communicates with a **MySQL database**, and all REST APIs were tested using **Postman**.

- A full stack approach offers a **complete end-to-end development perspective**, allowing seamless integration between the user interface, business logic, and data storage.

❖ Importance and Advantages

1. **Holistic Understanding:** Full stack developers have end-to-end knowledge of software development. This helps in better problem-solving and architectural decisions.

2. **Cost Efficiency**: Organizations can employ fewer developers when individuals can handle multiple layers of the development process.

3. **Faster Development and Troubleshooting**: With one person or a small team handling both sides of development, projects can be built and debugged more quickly.

4. **Career Opportunities**: Due to their broad skill set, full stack developers are in high demand across startups and large enterprises.

❖ Real-World Applications

- **E-commerce platforms** (e.g., Amazon, Flipkart): Require dynamic product listings, user authentication, and secure payments.

- **Social Media Apps** (e.g., Instagram, Twitter): Need a robust backend to manage user data and a responsive frontend to display feeds and notifications.

- **Content Management Systems** (e.g., WordPress, Ghost): Require CRUD operations for posts, user roles, and data storage.

- **Portfolio Websites**: Full stack knowledge allows individuals to host and showcase their own work, integrated with contact forms and blogs.

Frontend Development

Frontend development refers to the creation and management of the visual elements of a website or web application that users interact with directly in their web browsers. It is a crucial component of full stack development, as it serves as the **interface between users and the underlying functionality of the system**.

The primary objective of frontend development is to ensure that the user interface (UI) is not only visually appealing but also intuitive, accessible, responsive, and functional. A good frontend enhances user experience (UX), promotes engagement, and contributes significantly to the overall success of a web application.

Modern frontend development is no longer limited to just structure and design—it now encompasses **dynamic content rendering, form validations, animations, responsive layouts**, and even interactions with backend services using APIs.

❖ Role of Frontend in a Web Application

In a typical web application, the frontend is responsible for:

- Presenting data received from the backend to the user.

- Capturing user actions like clicks, form submissions, or keystrokes.
- Sending requests to the backend using APIs.
- Providing real-time feedback and visual responses.
- Adapting layout and elements to various screen sizes and devices.

This layer defines the overall **look, feel, and usability** of a website. A poorly designed frontend, regardless of how powerful the backend is, will deter users.

❖ **Technologies Used in Frontend Development**

Frontend development involves three core technologies:

HTML – HyperText Markup Language :

HTML is the foundation of all web content. It defines the structure of a webpage using tags. It is not a programming language but a markup language, which means it is used to annotate and organize content.

Example:

```
<!DOCTYPE html>
<html>
<head><title>My Page</title></head>
<body>
  <h1>Welcome to My Website</h1>
  <p>This is a sample paragraph.</p>
</body>
</html>
```

CSS – Cascading **S**tyle **S**heets :

CSS is used to style the content structured by HTML. While HTML provides the skeleton of a webpage, CSS adds design, colors, spacing, fonts, and layout. It separates content from presentation, making it easier to maintain and update the appearance of a site.

CSS follows a rule-based syntax where selectors target HTML elements and define visual properties. For example, using body { background-color: lightgray; } will change the page's background color. The layout is controlled using properties like margin, padding, display, and position. Advanced techniques such as Flexbox and CSS Grid enable developers to create complex, responsive layouts with ease.

Responsive styling can be achieved through media queries, which apply different styles depending on the screen width, resolution, or device orientation. Additionally, CSS can handle animations and transitions to enhance interactivity and provide feedback to user actions, such as hovering over a button.

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Example :

```css
body {
  background-color: #f0f0f0;
  font-family: Arial, sans-serif;
}
```

CSS enables:

- Layouts: grid, columns, positioning.
- Colors, fonts, and animations.
- Device adaptability (responsive design).

- CSS follows the **cascade principle**, where rules are applied based on priority and specificity.
- The **box model** governs the spacing and sizing of elements.
- **Responsive design** ensures that content looks good on all screen sizes, using techniques like media queries.
- CSS can be written inline, internally (within <style>), or externally (in .css files).

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JavaScript – Client-Side Scripting :

JavaScript is a dynamic programming language that allows developers to create interactive features on web pages. Unlike HTML and CSS, which are declarative and concerned with content and style respectively, JavaScript is procedural and event-driven, handling logic and user interactions in real time.

It is used to build features such as image sliders, form validations, pop-up alerts, and dynamic content updates. JavaScript can access and manipulate HTML elements using the Document Object Model (DOM). For instance, **document.getElementById("btn").style.display = "none";** hides a button when clicked.

JavaScript is also essential for communicating with back-end servers using APIs. With the introduction of ES6 (ECMAScript 2015) and later versions, the language now supports modern programming constructs such as arrow functions, classes, modules, and asynchronous operations through promises and async/await.

JavaScript forms the foundation of many modern frontend frameworks and libraries such as React, Angular, and Vue, making it indispensable in web development.

Features:

- Manipulate HTML and CSS through the DOM
- Validate forms before submission
- Handle events like clicks, scrolls, and keystrokes
- Make asynchronous API calls (AJAX, Fetch)

  Example :

```javascript
document.querySelector("#btn").addEventListener("click", () => {
  alert("Button Clicked!");
});
```

- JavaScript follows the **event-driven** and **asynchronous** programming models.
- It runs in the browser and responds to user interactions in real-time.
- Modern JavaScript (ES6 and beyond) introduces features like arrow functions, modules, classes, and promises.

JS can communicate with servers using technologies like **AJAX**, **Fetch API**, or **Axios**.

Modern Frontend Practices and Concepts :

Responsive Web Design (RWD) :-

Responsive Web Design (RWD) is an approach aimed at creating websites that adapt seamlessly to different screen sizes and devices. With the growing diversity in device types—smartphones, tablets, laptops, desktops—a one-size-fits-all layout no longer meets user expectations.

A responsive design ensures that content is readable, images scale properly, and navigation remains user-friendly across all screen resolutions. This is typically achieved through CSS media queries that apply styles conditionally based on screen width or device type. For example, a three-column layout on a desktop might become a single-column layout on a mobile device.

In addition to fluid grids and flexible images, RWD promotes a mobile-first design strategy, which prioritizes small screen experiences before enhancing them for larger displays. This not only improves usability but also contributes to better performance and faster loading times on mobile networks.

Responsive web design is essential in modern web development because it improves accessibility,

enhances SEO rankings, and provides a consistent experience across all platforms.

Responsive web design ensures that a website adapts seamlessly across different screen sizes and orientations. This is achieved using:

- Fluid Grids: Layouts defined in percentages rather than pixels
- Media Queries: Apply specific CSS based on device characteristics
- Flexible Images: Images scale depending on screen size

Accessibility (a11y) :-

Web accessibility ensures that websites are usable by people with various disabilities. This includes:

- Screen reader compatibility
- Keyboard navigation
- Color contrast
- Use of ARIA (Accessible Rich Internet Applications) attribute

Performance Optimization :-

Web performance affects user retention. Techniques include:

- Minifying CSS and JS files

- Lazy loading images
- Caching
- Using Content Delivery Networks (CDNs)

Tools and Frameworks :

CSS Frameworks

- **Bootstrap**: Offers prebuilt UI components
- **Tailwind CSS**: Utility-first approach for custom designs

Code Organization

Frontend code is organized into folders for better scalability:

```bash
/project
├── index.html
├── /css
│   └── styles.css
├── /js
│   └── script.js
├── /assets
    └── images
```

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➢ Backend Development

Backend development refers to the **server-side logic** of a web application — the part that handles **business logic, user authentication, data persistence**, and all communication between the frontend and database systems. While users interact only with the frontend, it is the backend that processes requests, enforces rules, and securely returns data.
In this project, the backend was implemented using **Java with Spring Boot**, a modern, production-ready web framework that simplifies REST API development, dependency injection, and security management. Data is stored in a **MySQL relational database**, and all backend logic follows a modular architecture based on **controllers, services, repositories, and entities**.

✅ Technologies Used in Backend

Layer	Technology
Language	Java
Framework	Spring Boot
ORM	JPA (Hibernate)
Database	MySQL
API Testing	Postman
Build Tool	Maven

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✅ Role of the Backend in the Web Application

- Handles all HTTP requests from the frontend (e.g., login, register, fetch products, submit orders).
- Validates and processes input data.
- Communicates with MySQL using JPA/Hibernate to perform CRUD operations.
- Manages authentication and authorization using JWT tokens or session-based logic.
- Serves static/dynamic content and returns structured JSON responses to the frontend.

✅ Spring Boot Overview

Spring Boot is a Java-based framework that allows developers to create stand-alone, production-grade applications easily. It comes with built-in support for:

- RESTful API creation
- Dependency Injection via Spring's powerful IoC container
- JPA for database mapping
- Auto-configuration for rapid development

Advantages:

- Lightweight and fast to boot
- Scalable and modular codebase
- Works seamlessly with MySQL
- Extensive community support

Database Integration

For storing, managing, and retrieving data efficiently, this project uses **MySQL**, a powerful relational database management system that is widely used in web applications.

MySQL – Relational Database System

MySQL organizes data in tables with rows and columns, where each table represents a specific entity like students, courses, attendance, etc. Tables are related through keys such as primary and foreign keys, enabling structured queries and relational integrity.

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Why MySQL?

- Robust support for transactions and relational data.
- Secure, scalable, and high-performance.
- Works well with PHP using MySQLi or PDO.
- Supports SQL (Structured Query Language) for querying and manipulating data.

Database Tables Used in the Project

- login (employeeid , name , designation , age , department , password)
- suggestions (id , title , current_scenario , suggested_scenario , implementation_cost , benefits , department , priority , status , created_date , updated_date)

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Creating a Table in MySQL

```sql
CREATE TABLE login (
    employeeid INT PRIMARY KEY,
    name VARCHAR(100),
    designation VARCHAR(100),
    age INT,
    department VARCHAR(100),
    password VARCHAR(255)
);
```

CRUD Operations in MySQL

Create (Insert)

```sql
INSERT INTO login (employeeid, name, designation, age, department, password)
VALUES (101, 'John Doe', 'Engineer', 30, 'Maintenance', 'john@123');
```

Read (Select)

```sql
SELECT * FROM login;
```

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Update

```
UPDATE login
SET designation = 'Senior Engineer', age = 31
WHERE employeeid = 101;
```

Delete

```
DELETE FROM login WHERE employeeid = 101;
```

Security Practices

MySQL support secure practices such as:

- Prepared statements to avoid SQL injection.
- Password hashing using password_hash() and password_verify().
- Session management for logged-in users.

Brief Mention: MongoDB

MongoDB is a NoSQL document-based database used in JavaScript-based backend stacks. It stores data in JSON-like format, making it suitable for flexible schema applications. While this project focuses on MySQL, understanding MongoDB is useful for future development in stacks like MERN (MongoDB, Express, React, Node).

INTRODUCTION

The success of any large-scale industrial organization depends significantly on the active involvement and innovative thinking of its workforce. In this context, the "Suggestion Scheme" project developed for RINL – Vizag Steel Plant aims to provide a digital platform that empowers employees to contribute constructive ideas and feedback directly to the management. This initiative fosters a participative culture and promotes continuous improvement within the organization.

The Suggestion Scheme is a web-based application designed to collect, manage, and evaluate employee suggestions in a structured and efficient manner. Developed using technologies such as JavaScript for the front-end and MySQL for the backend (via XAMPP server), the system allows employees to log in using their credentials, view their personal details, and submit suggestions through an intuitive interface. Each suggestion is stored in a secure database, enabling easy access and review by the concerned departments.

The primary objective of this project is to streamline the process of receiving suggestions, enhance transparency, and ensure that valuable insights from employees at all levels are not overlooked. By leveraging digital tools, the system reduces paperwork, improves data accuracy, and encourages a culture of innovation.

The implementation of such a suggestion mechanism not only improves operational efficiency but also enhances employee morale and engagement by recognizing their role in the plant's progress. This project stands as a step toward digital transformation and inclusive growth at RINL – Vizag Steel Plant.

OBJECTIVES

The main objectives of the Suggestion Scheme Web Application are as follows:

1. **To provide a centralized digital platform** for employees to submit suggestions aimed at improving plant operations, safety, productivity, and efficiency**.**

2. **To promote employee participation and innovation** by encouraging the workforce to contribute ideas and solutions for organizational betterment.

3. **To streamline the suggestion collection and review process** by replacing manual methods with an automated, database-driven system that ensures better tracking and management.

4. **To enhance transparency and accountability** in handling employee feedback by maintaining a structured system for storing, evaluating, and acting upon suggestions**.**

5. **To reduce administrative workload and paperwork** through the implementation of a web-based application, improving data accessibility and reporting**.**

6. **To support continuous improvement and digital transformation** initiatives at RINL by integrating employee insights into the organizational decision-making process.

KEY FEATURES OF THE SYSTEM

The **Suggestion Scheme Web Application** developed for RINL – Vizag Steel Plant includes a range of practical features to streamline the employee suggestion process and enhance operational efficiency. It offers employees and administrators a seamless and user-friendly experience with the following capabilities:

• Log in securely using unique employee credentials.

• View personal details such as name, designation, department, and age upon successful login.

• Submit suggestions directly through an interactive and easy-to-use suggestion form.

• Store all suggestions in a centralized MySQL database for efficient management and review.

• Ensure secure backend connectivity using XAMPP for local server support.

Each feature is built with simplicity and functionality in mind, allowing even non-technical users to navigate and use the system effectively.

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TECHNOLOGY STACK

Frontend: *HTML, CSS, JavaScript.*



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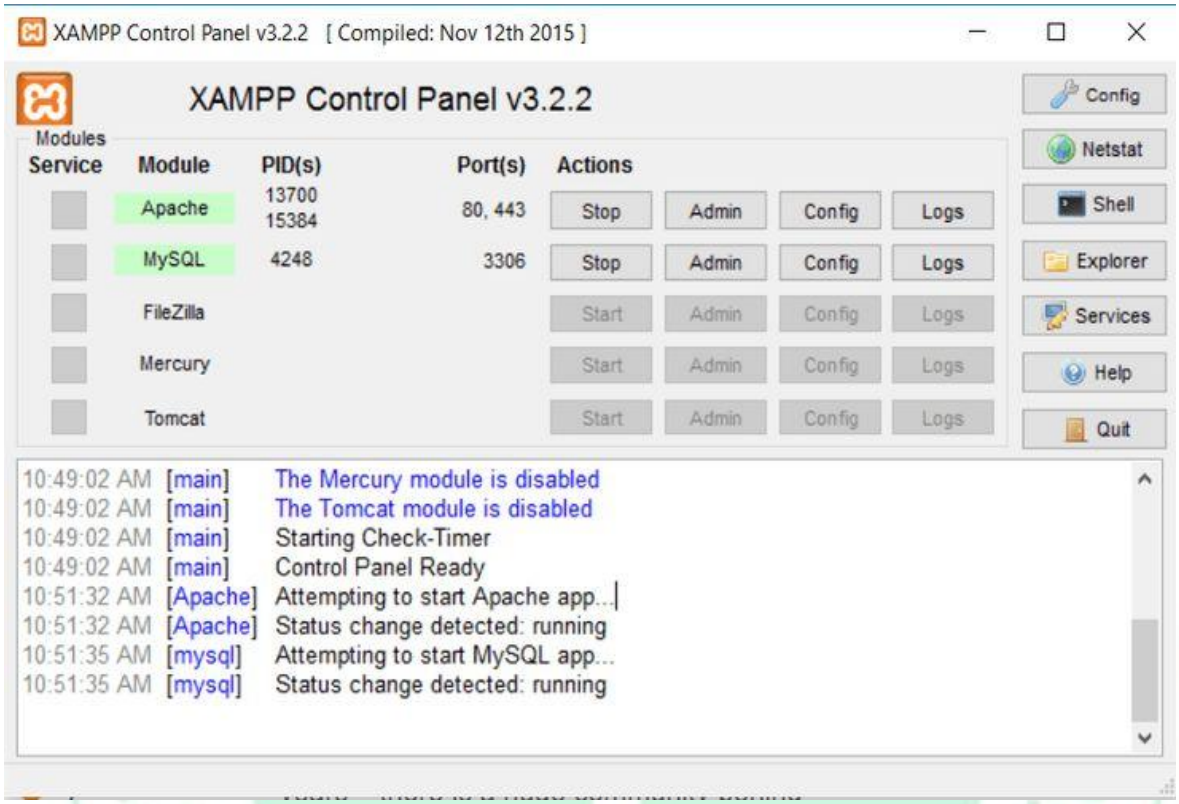
Backend: ***Java with Spring Boot***

- *RESTful APIs using @RestController*
- *Security handled with **Spring Security***
- *Project dependencies managed using **Maven***

Database: *MySQL.*

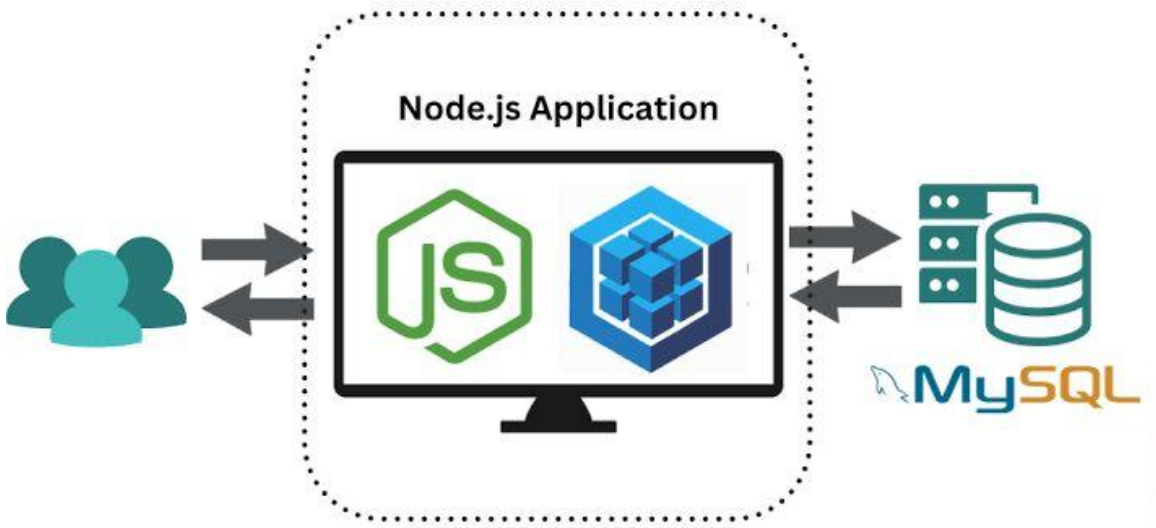


Local Server: *XAMPP (Apache + MySQL)*



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SYSTEM ARCHITECTURE



Client (Browser with JSP UI)

⇅

Spring Boot REST Application (Apache Tomcat Embedded)

⇅

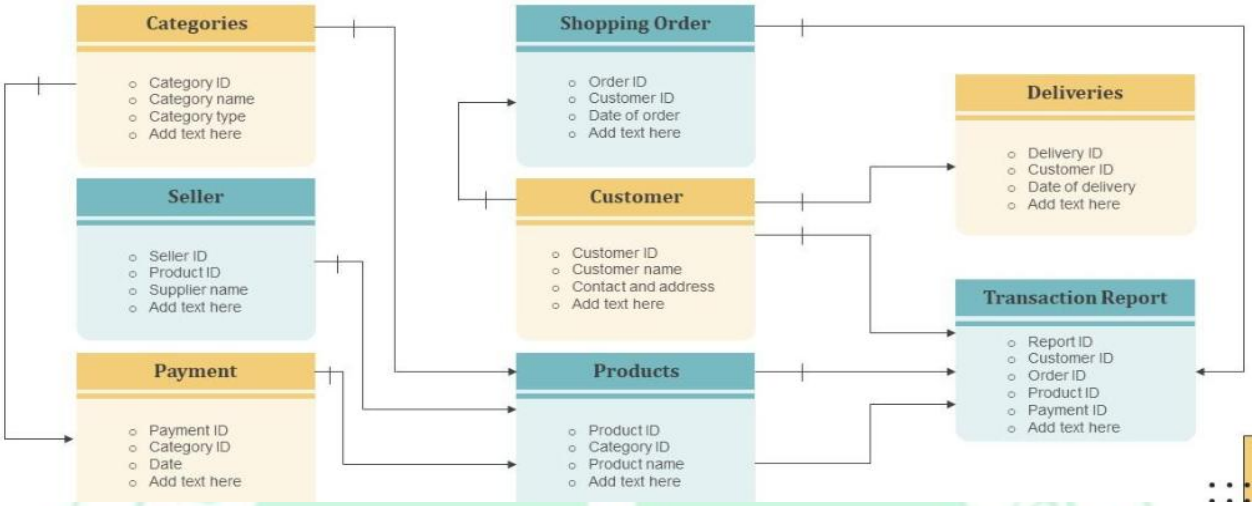
MySQL Database

The system follows a **layered MVC architecture**:

- The **frontend** (JSP + HTML/CSS/JS) sends requests via forms and AJAX.
- The **controller layer** in Spring Boot handles HTTP requests.
- The **service layer** processes business logic.
- The **repository layer** communicates with the **MySQL** database.

This structure ensures **separation of concerns**, better maintainability, and robust backend logic handling.

ER DIAGRAM



Entities and Their Attributes

1.Employee

- **employee_id (Primary Key):**_Unique identifier for each employee
- **name:** Full name of the employee
- **designation:** Job title or role
- **age:** Age of the employee
- **department:** Department in which the employee works
- **password:** Secure login credential (stored hashed)

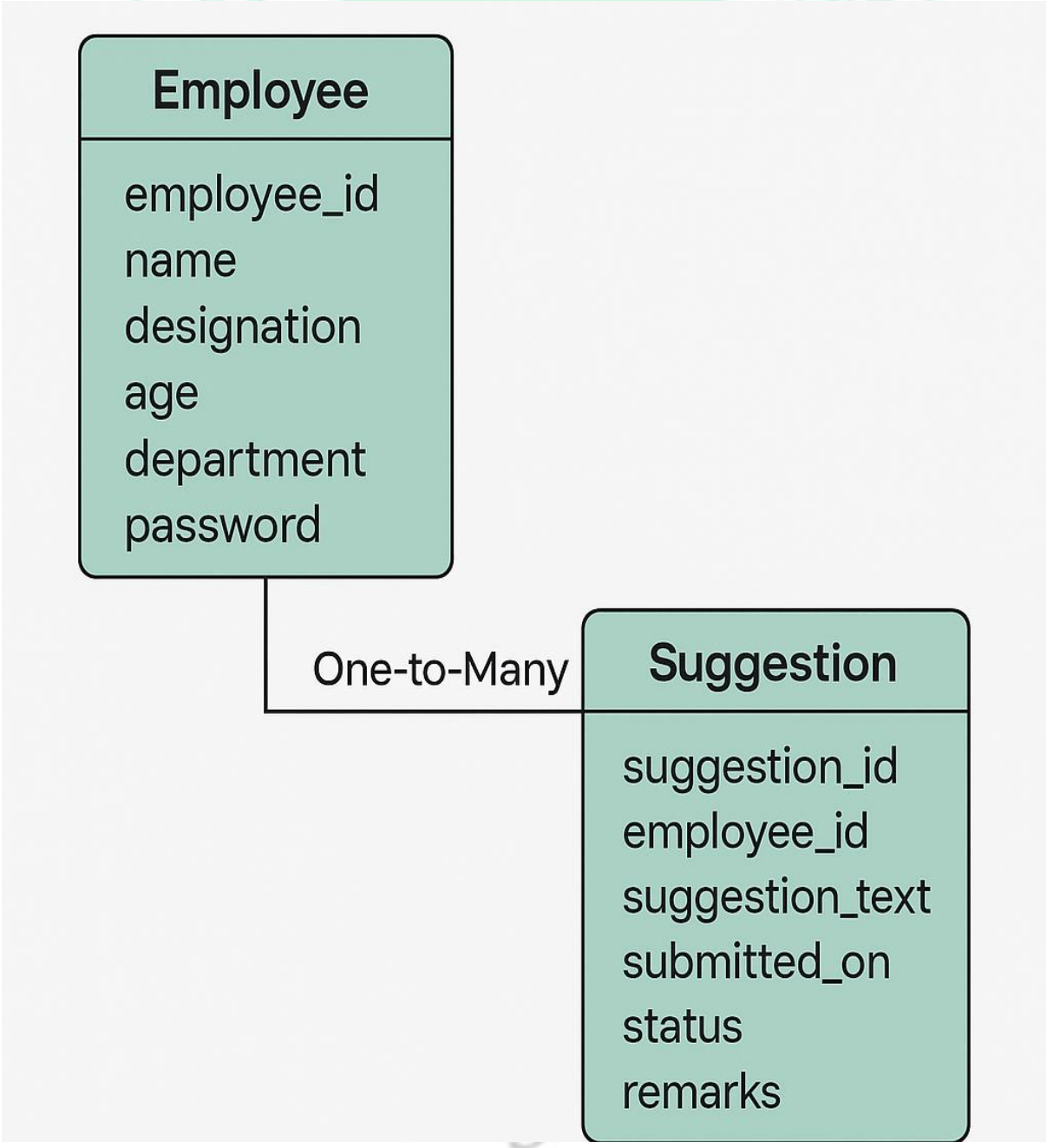
2. Suggestion

- **suggestion_id (Primary Key):** Unique identifier for each suggestion
- **employee_id (Foreign Key → Employee.employee_id):** Link to the employee who submitted the suggestion
- **suggestion_text:** Content of the suggestion
- **submitted_on:** Date and time of submission
- **status:** Status of the suggestion (e.g., pending, reviewed, accepted, rejected)

Relationships

Employee – Suggestion

- One-to-Many relationship
- One employee can submit multiple suggestions
- Each suggestion is submitted by only one employee



Entity	Relationship Type	With Entity
User	One-to-Many	Address
User	One-to-One	Cart
User	One-to-Many	Order, Review, Transaction
User	Many-to-Many	Coupon
User	One-to-One	Wishlist, Verification- Code
Cart	One-to-Many	Cart Item
Cart Item	Many-to-One	Product
Product	Many-to-One	Seller, Category
Product	One-to-Many	Review
Order	One-to-Many	Order Item
Order	Many-to-One	Address, User
Transaction	Many-to-One	User, Seller
Seller	One-to-Many	Product, Transaction
Seller	One-to-One	Seller Report, Address
Wishlist	Many-to-Many	Product

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MODULES

1. **Login Module**
   • Authenticates employee credentials using JavaScript and MySQL through the XAMPP server.

2. **Dashboard Module**
   • Displays employee details such as name, designation, department, and age after successful login.

3. **Suggestion Submission Module**
   • Allows employees to enter and submit suggestions using a simple and interactive form interface.

4. **Database Interaction Module**
   • Handles all backend data operations such as storing and retrieving suggestions using MySQL queries.

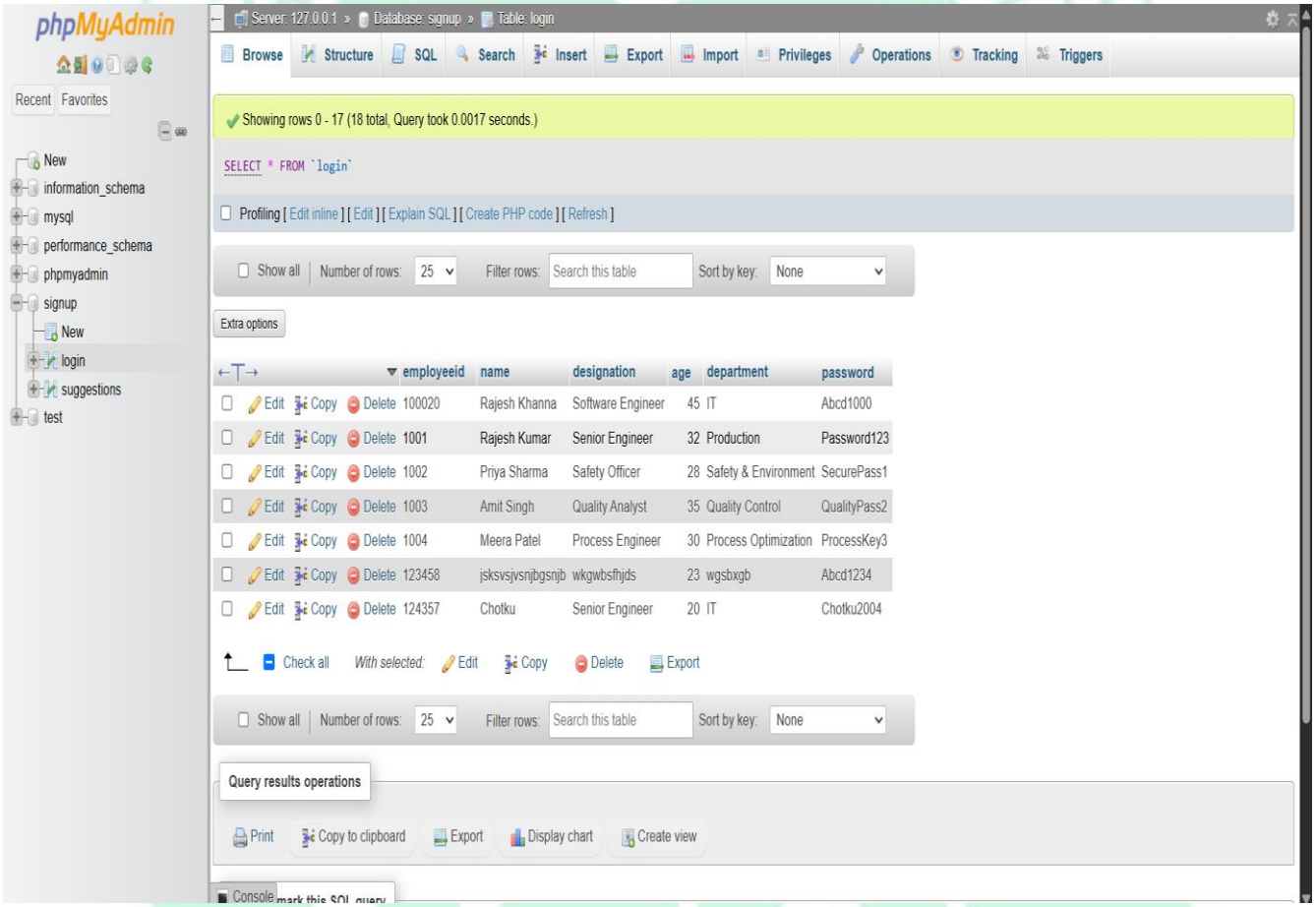
5. **Logout Module**
   • Safely terminates the user session and redirects to the login page to ensure security.

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DATABASE DESIGN

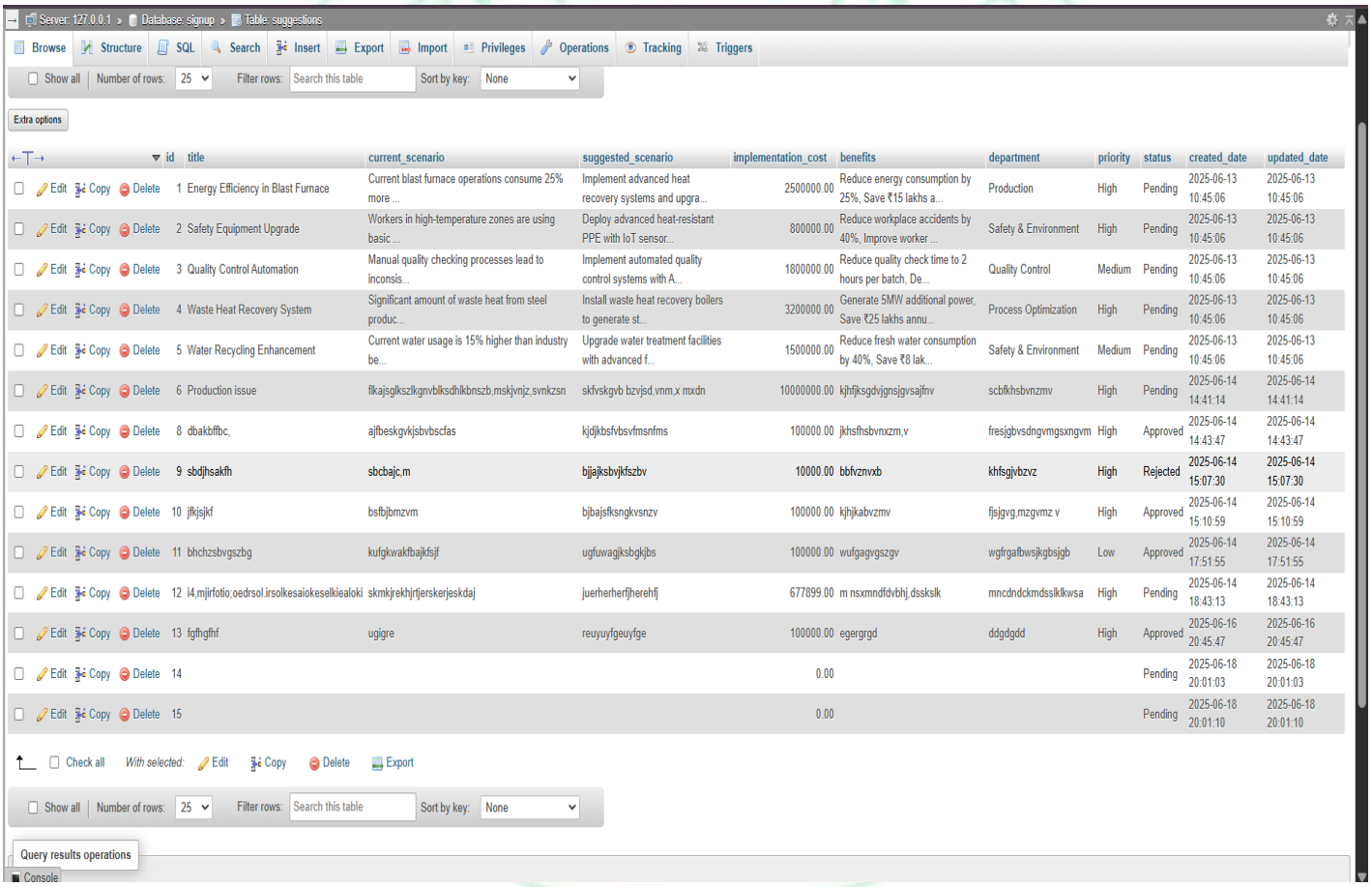
Tables:

- **login (employeeid , name , designation , age , department , password)**



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- **suggestions (id , title , current_scenario , suggested_scenario , implementation_cost , benefits , department , priority , status , created_date , updated_date)**



Server: 127.0.0.1 » Database: signup » Table: suggestions

Browse | Structure | SQL | Search | Insert | Export | Import | Privileges | Operations | Tracking | Triggers

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	id	title	current_scenario	suggested_scenario	implementation_cost	benefits	department	priority	status	created_date	updated_date
Edit Copy Delete	1	Energy Efficiency in Blast Furnace	Current blast furnace operations consume 25% more ...	Implement advanced heat recovery systems and upgra...	2500000.00	Reduce energy consumption by 25%, Save ₹15 lakhs a...	Production	High	Pending	2025-06-13 10:45:06	2025-06-13 10:45:06
Edit Copy Delete	2	Safety Equipment Upgrade	Workers in high-temperature zones are using basic ...	Deploy advanced heat-resistant PPE with IoT sensor...	800000.00	Reduce workplace accidents by 40%, Improve worker ...	Safety & Environment	High	Pending	2025-06-13 10:45:06	2025-06-13 10:45:06
Edit Copy Delete	3	Quality Control Automation	Manual quality checking processes lead to inconsis...	Implement automated quality control systems with A...	1800000.00	Reduce quality check time to 2 hours per batch, De...	Quality Control	Medium	Pending	2025-06-13 10:45:06	2025-06-13 10:45:06
Edit Copy Delete	4	Waste Heat Recovery System	Significant amount of waste heat from steel produc...	Install waste heat recovery boilers to generate st...	3200000.00	Generate 5MW additional power, Save ₹25 lakhs annu...	Process Optimization	High	Pending	2025-06-13 10:45:06	2025-06-13 10:45:06
Edit Copy Delete	5	Water Recycling Enhancement	Current water usage is 15% higher than industry be...	Upgrade water treatment facilities with advanced f...	1500000.00	Reduce fresh water consumption by 40%, Save ₹8 lak...	Safety & Environment	Medium	Pending	2025-06-13 10:45:06	2025-06-13 10:45:06
Edit Copy Delete	6	Production issue	flkajsglkszlkgnvblksdhlkbnszb,mskjvnjz,svnkzsn	skfvskgvb bzvjsd,vnm,x mxdn	10000000.00	kjhfjksgdvjgnsjgvsajfnv	scbfkhsbvnzmv	High	Pending	2025-06-14 14:41:14	2025-06-14 14:41:14
Edit Copy Delete	8	dbakbffbc,	ajfbeskgvkjsbvbscfas	kjdjkbsfvbsvfmsnfms	100000.00	jkhsfhsbvnxzm,v	fresjgbvsdngvmgsxngvm	High	Approved	2025-06-14 14:43:47	2025-06-14 14:43:47
Edit Copy Delete	9	sbdjhsakfh	sbcbajc,m	bjjajksbvjkfszbv	10000.00	bbfvznvxb	khfsgjvbzvz	High	Rejected	2025-06-14 15:07:30	2025-06-14 15:07:30
Edit Copy Delete	10	jfkjsjkf	bsfbjbmzvm	bjbajsfksngkvsnzv	100000.00	kjhjkabvzmv	fjsjgvg,mzgvmz v	High	Approved	2025-06-14 15:10:59	2025-06-14 15:10:59
Edit Copy Delete	11	bhchzsbvgszbg	kufgkwakfbajkfsjf	ugfuwagjksbgkjbs	100000.00	wufgagvgszgv	wgfrgafbwsjkgbsjgb	Low	Approved	2025-06-14 17:51:55	2025-06-14 17:51:55
Edit Copy Delete	12	i4,mjirfotio;oedrsol.irsolkesaiokeselkiealoki	skmkjrekhjrtjerskerjeskdaj	juerherherfjherehfj	677899.00	m nsxmndfdvbhj,dsskslk	mncdndckmdsslklkwsa	High	Pending	2025-06-14 18:43:13	2025-06-14 18:43:13
Edit Copy Delete	13	fgfhgfhf	ugigre	reuyuyfgeuyfge	100000.00	egergrgd	ddgdgdd	High	Approved	2025-06-16 20:45:47	2025-06-16 20:45:47
Edit Copy Delete	14				0.00				Pending	2025-06-18 20:01:03	2025-06-18 20:01:03
Edit Copy Delete	15				0.00				Pending	2025-06-18 20:01:10	2025-06-18 20:01:10

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Console

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CODE SNIPPETS

Database Connection (db.js):

```
const mysql = require('mysql');

let db;

function handleConnection() {
 db = mysql.createConnection({
   host: "localhost",
   user: "root",
   password: "",
   database: "signup"
 });

 db.connect((err) => {
   if (err) {
     console.error("❌ DB connection failed. Retrying in 2 seconds...");
     setTimeout(handleConnection, 2000);
   } else {
     console.log("✅ DB connected");
   }
 });
```

```
  db.on("error", (err) => {
    if (err.code === "PROTOCOL_CONNECTION_LOST") {
      handleConnection();
    } else {
      throw err;
    }
  });
}

handleConnection();

module.exports = db;
```

Login Route (Server.js):

```
const express = require("express");
const cors = require("cors");
const db = require("./db");

const app = express();
app.use(cors());
app.use(express.json());

// ✅ LOGIN: Match employeeid and password
app.post("/login", (req, res) => {
  const { employeeid, password } = req.body;

  const sql = `
    SELECT employeeid, name, designation, age, department
    FROM login
    WHERE TRIM(employeeid) = ? AND TRIM(password) = ?
  `;

  db.query(sql, [employeeid, password], (err, result) => {
    if (err) return res.status(500).json({ message: "Database error" });

    if (result.length > 0) {
      res.status(200).json({ status: "Success", user: result[0] });
    } else {
      res.status(401).json({ status: "Failed", message: "Invalid credentials" });
    }
  });
});
```

Frontend Login Module – React (Login.js):

```
import React, { useState } from 'react';
import { Link, useNavigate } from 'react-router-dom';
import axios from 'axios';
import Validation from './LoginValidation';

function Login() {
  const [values, setValues] = useState({ employeeid: '', password: '' });
  const [errors, setErrors] = useState({});
  const [loginError, setLoginError] = useState('');
  const [toastVisible, setToastVisible] = useState(false);
  const [toastMessage, setToastMessage] = useState('');
  const navigate = useNavigate();

  const handleInput = (e) => {
    setValues(prev => ({ ...prev, [e.target.name]: e.target.value }));
    setLoginError('');
  };
```

```
const handleSubmit = (e) => {
  e.preventDefault();
  const validationErrors = Validation(values);
  setErrors(validationErrors);

  if (Object.keys(validationErrors).length === 0) {
    axios.post('http://localhost:8081/login', values)
      .then(res => {
        if (res.data.status === "Success") {
          localStorage.setItem("employee", JSON.stringify(res.data.user));
          setToastMessage("✅ Logged in successfully");
          setToastVisible(true);
          setTimeout(() => {
            setToastVisible(false);
            navigate('/home', { state: { user: res.data.user } });
          }, 2000);
        } else {
          setLoginError("❌ Invalid employee ID or password");
        }
      })
```

```
      .catch(() => setLoginError("❌ Server error. Please try again later."));
    }
  };

  return (
    <div className="login-page">
      {/* Login Form UI */}
      <form onSubmit={handleSubmit}>
        <input type="number" name="employeeid" onChange={handleInput} placeholder="Enter Employee
        <input type="password" name="password" onChange={handleInput} placeholder="Enter Password"
        <button type="submit">Log in</button>
      </form>
      {loginError && <span>{loginError}</span>}
    </div>
  );
}

export default Login;
```

Frontend Suggestion Submission Module – React (Home.js)

```
import React, { useState } from 'react';
import { useLocation } from 'react-router-dom';
import axios from 'axios';

function Home() {
  const location = useLocation();
  const routeUser = location.state?.user;
  const localUser = JSON.parse(localStorage.getItem("employee"));
  const user = routeUser || localUser || {};

  const [form, setForm] = useState({
    title: '',
    current_scenario: '',
    suggested_scenario: '',
    implementation_cost: '',
    benefits: '',
    department: '',
    priority: '',
    status: 'Pending'
  });
```

```
const [submitMessage, setSubmitMessage] = useState('');
const [toastVisible, setToastVisible] = useState(false);

const handleChange = (e) => {
  setForm(prev => ({ ...prev, [e.target.name]: e.target.value }));
};

const handleSubmit = (e) => {
  e.preventDefault();
  axios.post("http://localhost:8081/submit-suggestion", form)
    .then(() => {
      setSubmitMessage("Suggestion submitted!");
      setToastVisible(true);
      setForm({
        title: '',
        current_scenario: '',
        suggested_scenario: '',
        implementation_cost: '',
        benefits: '',
        department: '',
        priority: '',
        status: 'Pending'
      });
```

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```
        setTimeout(() => setToastVisible(false), 3000);
      })
      .catch(() => {
        setSubmitMessage("Submission failed");
        setToastVisible(true);
        setTimeout(() => setToastVisible(false), 3000);
      });
  };

  return (
    <div>
      <h1>Employee Suggestion Form</h1>
      <p><strong>Name:</strong> {user.name}</p>
      <p><strong>Designation:</strong> {user.designation}</p>
      <form onSubmit={handleSubmit}>
        <input name="title" value={form.title} onChange={handleChange} placeholder="Title" />
        <textarea name="current_scenario" value={form.current_scenario} onChange={handleChange}
placeholder="Current Scenario" />
        <textarea name="suggested_scenario" value={form.suggested_scenario} onChange={handleChange}
placeholder="Suggested Scenario" />
        <input name="implementation_cost" value={form.implementation_cost} onChange={handleChange}
placeholder="Implementation Cost" />
        <textarea name="benefits" value={form.benefits} onChange={handleChange} placeholder="Benefits" />
        <input name="department" value={form.department} onChange={handleChange} placeholder="Department" />
        <select name="priority" value={form.priority} onChange={handleChange}>
          <option value="">Select Priority</option>
          <option value="High">High</option>
          <option value="Medium">Medium</option>
          <option value="Low">Low</option>
        </select>
        <button type="submit">Submit</button>
      </form>
      {toastVisible && <div>{submitMessage}</div>}
    </div>
  );
}

export default Home;
```

Frontend Routing Module – React (App.js)

```
import React from 'react';
import Login from './Login';
import {BrowserRouter, Routes, Route} from 'react-router-dom'
import Signup from './Signup'
import Home from './Home';


function App() {
  return (
    <BrowserRouter>
    <Routes>
      <Route path='/' element={<Login />}></Route>
      <Route path='/signup' element={<Signup />}></Route>
      <Route path='/home' element={<Home />}></Route>
    </Routes>
    </BrowserRouter>
  );
}

export default App;
```

STYLESHEETS AND UI DESIGN

login.css: adds glowing hover effects and label animations for a smooth, interactive UI.

```
.login-page {
  position: relative;
  min-height: 100vh;
  display: flex;
  flex-direction: column;
  overflow-y: auto;
}

.login-bg {
  display: flex;
  justify-content: center;
  align-items: center;
  padding: 60px 20px;
  min-height: calc(100vh - 100px);
  position: relative;
}
```

```
.login-bg::before {
  content: "";
  position: absolute;
  top: 0; left: 0;
  width: 100%;
  height: 100%;
  background: url('./assets/steeel-bg.jpg') no-repeat center center/cover;
  filter: blur(6px);
  z-index: 0;
}

/* Form Styling */
.login-form {
  position: relative;
  z-index: 1;
  max-width: 450px;
  padding: 2.5rem;
  background-image: linear-gradient(334deg, #071d57, #2d4b51, #337d46, #14b12e);
  border-radius: 1rem;
  box-shadow: 0 0 30px rgba(0, 0, 0, 0.25);
  color: white;
}
```

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```
.login-form input,
.login-form .form-control {
  background-color: rgba(255, 255, 255, 0.95);
  color: #000;
  border: none;
}

.login-form .btn-success {
  background-color: #1ac959;
  border-color: #1ac959;
}

.login-form .btn-success:hover {
  background-color: #159e43;
  border-color: #159e43;
}
```

```
/* Navbar Styling */
.navbar-custom {
  display: flex;
  align-items: center;
  padding: 1rem 2rem;
  background-color: rgb(25,135,84);
  box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
}
.navbar-logo {
  height: 80px;
  margin-right: 20px;
}
.navbar-title {
  font-size: 4rem;
  font-family: "Alike Angular", serif;
  font-weight: bold;
  color: white;
  padding-left: 28.5%;
}
/* Title Banner */
.title-banner {
  text-align: center;
  font-size: 50px;
  font-weight: 600;
  margin: 25px auto 0;
  color: rgb(40, 174, 112);
  text-shadow: 1px 1px 2px #000;
}
```

Suggestion.css: Provides a clean, structured layout for the suggestion form with responsive design, styled input fields, and toast notifications for user feedback.

```
return (
  <div style={{ padding: "20px", fontFamily: "Arial" }}>
    <h1 style={{ textAlign: "center" }}>Employee Suggestion Form</h1>

    {/* ✅ Employee Info Card */}
    <div style={{
      backgroundColor: "#f0f4ff",
      padding: "15px",
      borderRadius: "10px",
      margin: "20px auto",
      maxWidth: "800px",
      boxShadow: "0 0 5px rgba(0,0,0,0.1)"
    }}>
      <h2>👋 Welcome {user.name || "Employee"}</h2>
      <p><strong>Employee ID:</strong> {user.employeeid || "N/A"}</p>
      <p><strong>Name:</strong> {user.name || "N/A"}</p>
      <p><strong>Designation:</strong> {user.designation || "N/A"}</p>
      <p><strong>Age:</strong> {user.age || "N/A"}</p>
      <p><strong>Department:</strong> {user.department || "N/A"}</p>
    </div>
```

```
  {/* ✅ Suggestion Form */}
  <form onSubmit={handleSubmit} style={{ maxWidth: "800px", margin: "auto" }}>
    <FormBlock title="Title" name="title" value={form.title} onChange={handleChange}
      placeholder="Enter a brief title for your suggestion" />
    <FormBlock title="Current Scenario" name="current_scenario" value={form.current_scenario}
      onChange={handleChange} placeholder="Describe the current process or issue" multiline />
    <FormBlock title="Suggested Scenario" name="suggested_scenario" value={form.suggested_scenario}
      onChange={handleChange} placeholder="Describe your proposed improvement or solution" multiline />
    <FormBlock title="Implementation Cost (₹)" name="implementation_cost"
      value={form.implementation_cost} onChange={handleChange}
      placeholder="Estimated cost in INR, e.g., 500000" />
    <FormBlock title="Benefits" name="benefits" value={form.benefits}
      onChange={handleChange} placeholder="List the expected benefits or advantages" multiline />
    <FormBlock title="Department" name="department" value={form.department}
      onChange={handleChange} placeholder="Department this suggestion is related to" />
    <SelectBlock title="Priority" name="priority" value={form.priority}
      onChange={handleChange} options={["High", "Medium", "Low"]} />
    <SelectBlock title="Status" name="status" value={form.status}
      onChange={handleChange} options={["Pending", "Approved", "Rejected"]} />
```

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```
      <button type="submit" style={{
        marginTop: "15px",
        padding: "10px 20px",
        fontSize: "16px",
        border: "1px solid black",
        borderRadius: "5px",
        cursor: "pointer"
      }}>
        Submit
      </button>
    </form>

    {/* ✅ Toast Popup */}
    <Toast message={submitMessage} visible={toastVisible} />
  </div>
);
```

```
const Toast = ({ message, visible }) => (
  <div style={{
    position: "fixed",
    bottom: "30px",
    right: "30px",
    backgroundColor: message.includes("failed") ? "#dc3545" : "#28a745",
    color: "white",
    padding: "12px 20px",
    borderRadius: "6px",
    boxShadow: "0 0 10px rgba(0,0,0,0.3)",
    fontSize: "16px",
    opacity: visible ? 1 : 0,
    transition: "opacity 0.5s ease-in-out",
    zIndex: 1000
  }}>
    ✅ {message}
  </div>
);
```

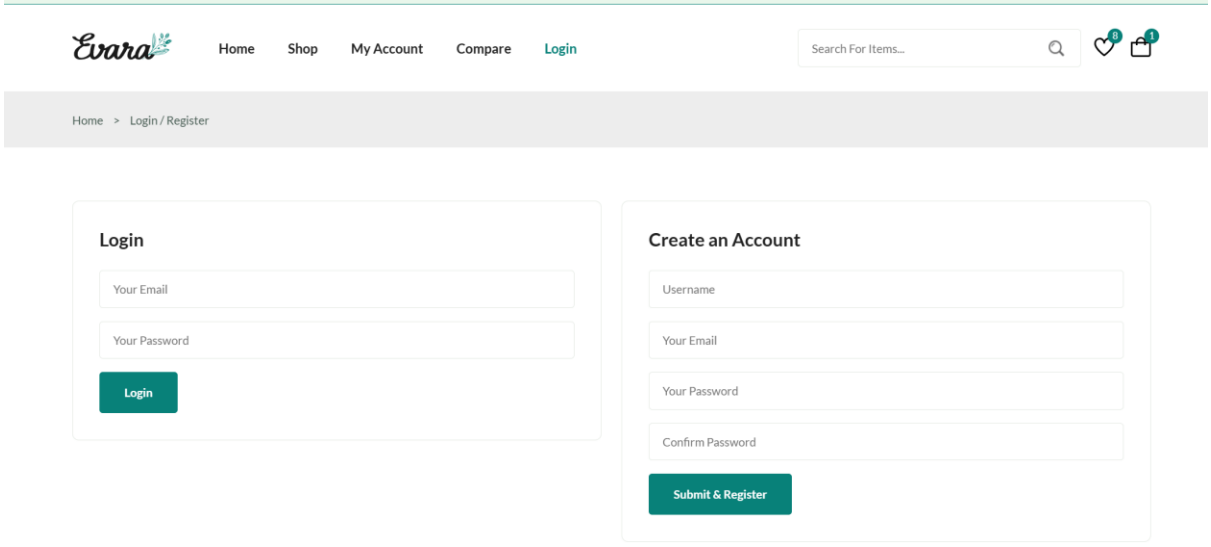
```
const FormBlock = ({ title, name, value, onChange, placeholder, multiline }) => (
  <div style={{ marginBottom: "15px" }}>
    <label><strong>{title}</strong></label>
    {multiline ? (
      <textarea
        name={name}
        rows="4"
        value={value}
        onChange={onChange}
        placeholder={placeholder}
        style={{ width: "100%", padding: "10px", borderRadius: "4px", border: "1px solid #ccc" }}
      />
    ) : (
      <input
        name={name}
        value={value}
        onChange={onChange}
        placeholder={placeholder}
        style={{ width: "100%", padding: "10px", borderRadius: "4px", border: "1px solid #ccc" }}
      />
    )}
  </div>
);
```

```
const SelectBlock = ({ title, name, value, onChange, options }) => (
  <div style={{ marginBottom: "15px" }}>
    <label><strong>{title}</strong></label>
    <select
      name={name}
      value={value}
      onChange={onChange}
      style={{ width: "100%", padding: "10px", borderRadius: "4px", border: "1px solid #ccc" }}
    >
      <option value="">Select {title}</option>
      {options.map((opt, idx) => (
        <option key={idx} value={opt}>{opt}</option>
      ))}
    </select>
  </div>
);
```

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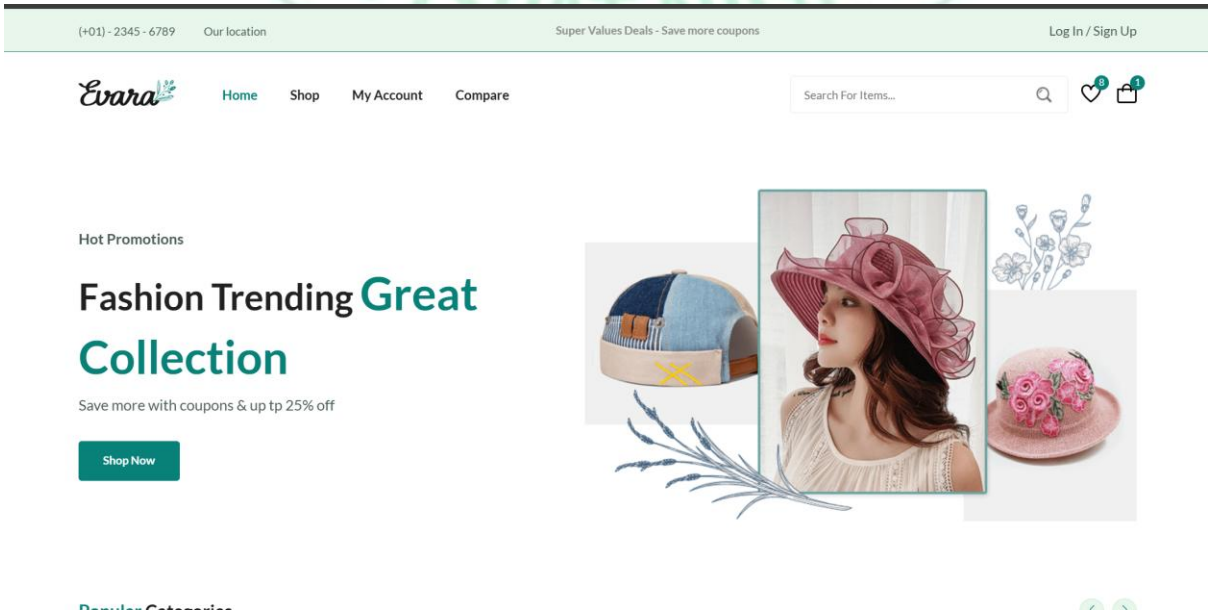
Interface Overview & Page layout

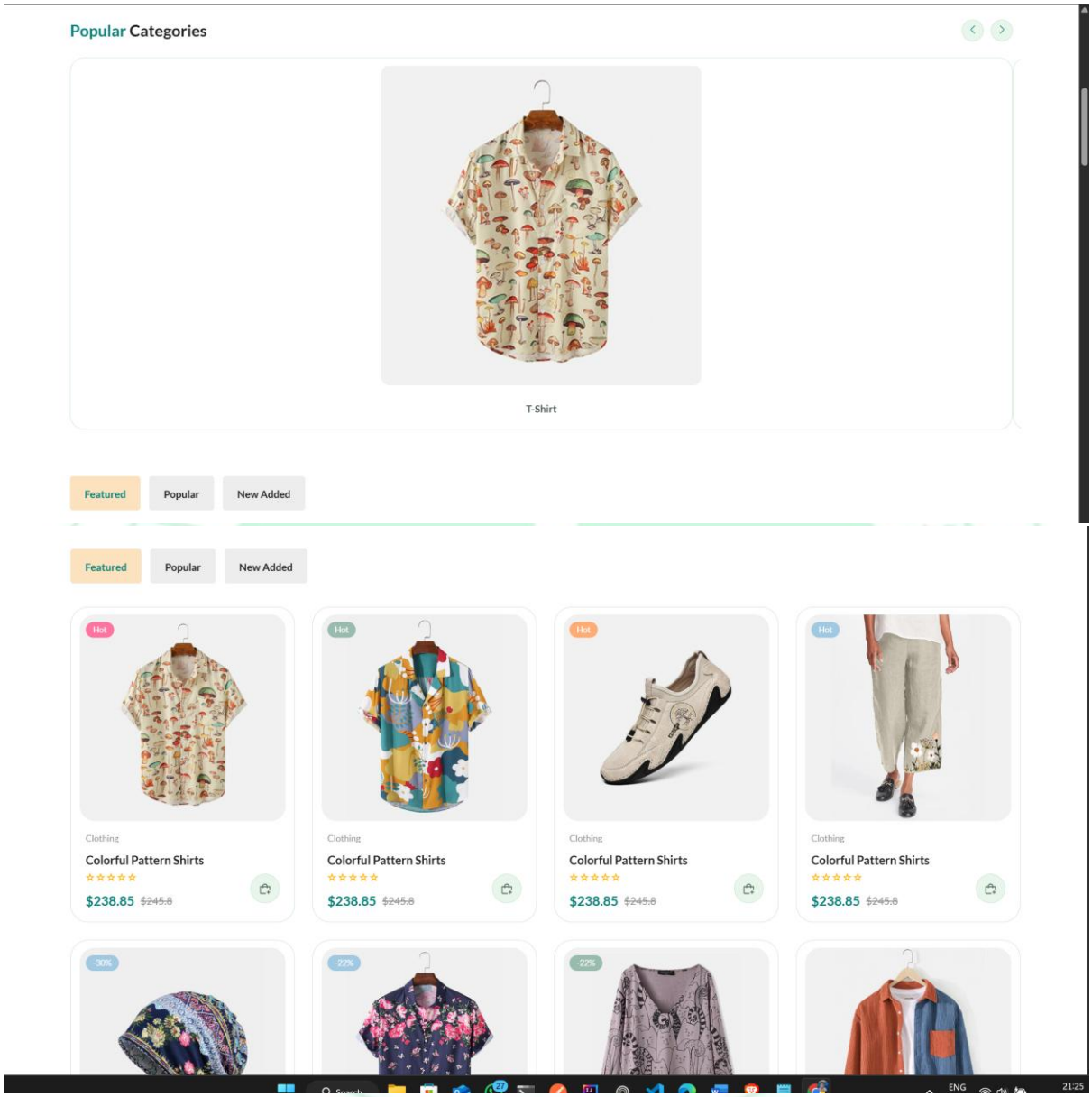
Login Page:



Suggestion Scheme Page:

Dashboard:





Compare Dashboard:

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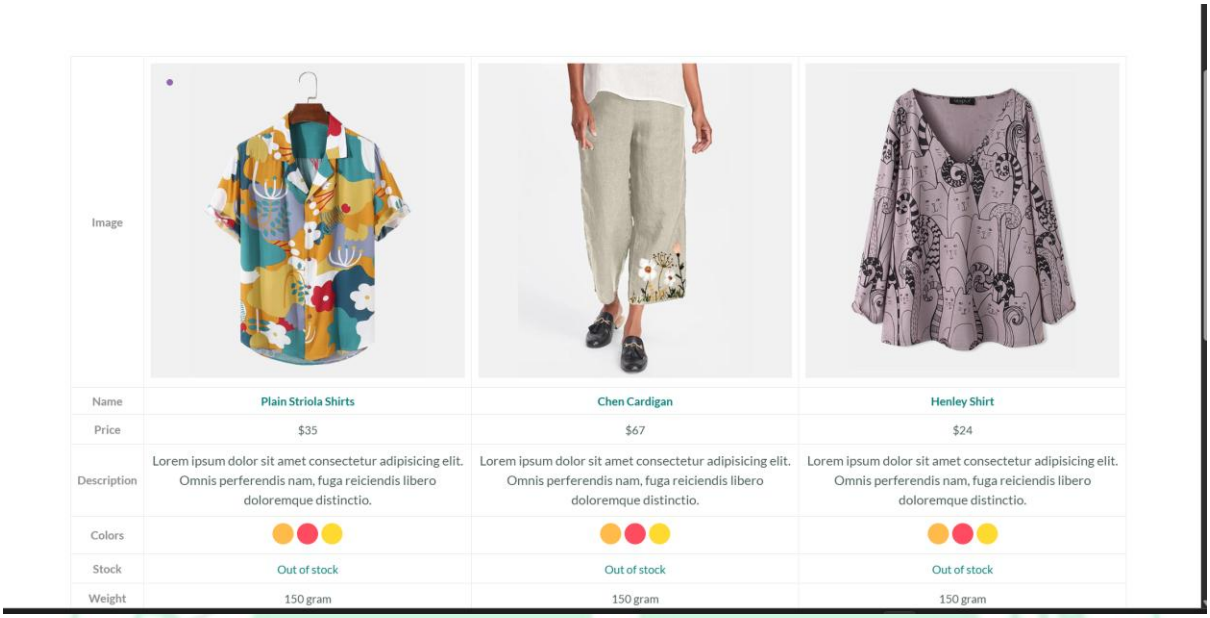


Image			
Name	Plain Striola Shirts	Chen Cardigan	Henley Shirt
Price	$35	$67	$24
Description	Lorem ipsum dolor sit amet consectetur adipisicing elit. Omnis perferendis nam, fuga reiciendis libero doloremque distinctio.	Lorem ipsum dolor sit amet consectetur adipisicing elit. Omnis perferendis nam, fuga reiciendis libero doloremque distinctio.	Lorem ipsum dolor sit amet consectetur adipisicing elit. Omnis perferendis nam, fuga reiciendis libero doloremque distinctio.
Colors			
Stock	Out of stock	Out of stock	Out of stock
Weight	150 gram	150 gram	150 gram

SECURITY CONSIDERATIONS

- **Credential-Based Login System**
  Employee access is restricted through a secure login mechanism using unique employee ID and password.
- **Password Masking and Field Validation**
  Passwords are masked on the frontend and validated to prevent empty or weak submissions.
- **Server-Side Verification**
  Backend verifies credentials using trimmed and validated inputs, rejecting malformed or invalid requests.

TESTING AND DEBUGGING

- Synchronizing frontend form submission with backend API responses using Axios.
- Debugging Node.js and Express routing issues during login and suggestion submission.
- Validating form inputs and ensuring error handling for invalid or missing fields.

- Testing localStorage-based session persistence and logout functionality.
- Ensuring responsive design and layout consistency across different screen sizes.
- Verifying toast notifications for both success and failure states during operations.

Tools Used

- **IntelliJ IDEA / VS Code** for development
- **Spring Boot** (backend)
- **MySQL** (database)
- **React + TypeScript + Redux Toolkit** (frontend)
- **JWT, Spring Security, Java Mail Sender**
- **Razorpay & Stripe** payment integrations

FUTURE SCOPE

- Admin Dashboard Integration.
- Mobile Application Support.

- Email Notifications.
- Multi-Language Support.

REFERENCES

- Bootstrap Documentation
- Spring Boot
- MDN Web Docs
- W3Schools Tutorials
- Stack Overflow
- YouTube
- MySQL Reference Manual
- Node.js Official Documentation
- Express.js Guide
- GeeksforGeeks

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CONCLUSION

The internship at **Vizag Steel Plant (RINL)** served as a valuable platform for gaining hands-on experience in **Full Stack Web Development**, blending academic knowledge with real-world application.

During this internship, we successfully conceptualized and developed a **secure, user-friendly, and responsive web-based Suggestion Scheme system**. The application empowers employees to share improvement ideas directly with management, thereby fostering a culture of innovation, participation, and continuous organizational improvement.

This system marks a major shift from traditional, manual suggestion handling to a **digitally streamlined process**. It significantly **reduces paperwork, minimizes human error**, and ensures better **data tracking and accountability**. Features such as **real-time feedback, form validation, and employee-specific suggestion history** enhance both user engagement and operational efficiency.

From a technical standpoint, the project strengthened our competencies in:

- **Frontend-backend integration** using **Node.js** and **React**
- **RESTful API design and implementation**
- **Session management** using browser storage
- **Database connectivity and CRUD operations** using **MySQL**
- **Security and form validation best practices**

These skills are essential for building scalable and maintainable web applications in any enterprise context.

In conclusion, the **Suggestion Scheme Web Application** not only meets its core objective of simplifying and digitizing idea submission but also lays the groundwork for future extensions such as:

- **Admin dashboards**
- **Automated notifications**
- **Suggestion analytics**
- **Integration with HR and performance systems**

This initiative is a significant step forward in RINL's journey toward **digital transformation and inclusive growth**.

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